Project 3: 555 Timer and 8-bit Shift Register

This project is a specific application of a very general skill: Understanding and working with compact, complex circuits and components. These compact circuits are typically called ICs (Integrated Circuits) and are very useful when prototyping and manufacturing larger electronics. Be aware that although we use a 555 timer and 8-bit Shift Register, these are **not** the most important concepts of this project: it is **understanding how to work with ICs in general**.

Objectives

- Understand the structure of ICs and how they can be used in design
- Learn to read IC schematic sheets
- Control frequency of a speaker using a 555 timer
- Control frequency of a series of LEDs with an 8 bit shift Register

Task Description

Summary:

In this project, you will work with a 555 timer, a specific integrated circuit (IC) which can be used as an oscillator. Since it oscillates the signal based on the signals it receives, we can change how frequently it oscillates.

Your task is to use this oscillating signal generated from the 555 chip with resistors and buttons to create a piano with 3 frequencies. For those interested, the circuit inside of a 555 IC can be found here. A sample schematic of a 555 timer circuit can be found on the next page.

What you will need:

- 1x Arduino (The Arduino is used strictly as a convenient 5V DC power and ground source for your breadboard rails)
- 1x 555 Timer IC
  - https://www.digikey.com/en/products/detail/texas-instruments/NE555P/277057
- 1x 8-bit Shift Register
  - https://www.digikey.com/en/products/detail/texas-instruments/SN74HC595N/277246
- 4x Pushbuttons
  - https://www.digikey.com/en/products/detail/cui-devices/TS02-66-55-BK-160-LCR-D/15634374

- 5x 10k Ω Resistors
- 4x 330 Ω Resistors
- 4x LEDs
- 1x 0.1 uF capacitor
- 1x 100 uF capacitor (Provided in today's lecture)

Part 1:

1. Successfully wire your 555 timer on your breadboard
2. Include push buttons in design (elements S1, S2, S3)
3. Manipulate the buttons so pressing them changes your piezo frequency.

Figure 1: 555 Timer Pin Layout

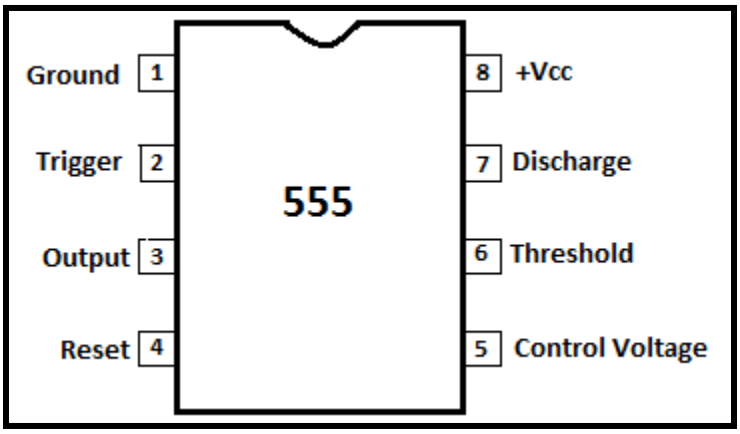
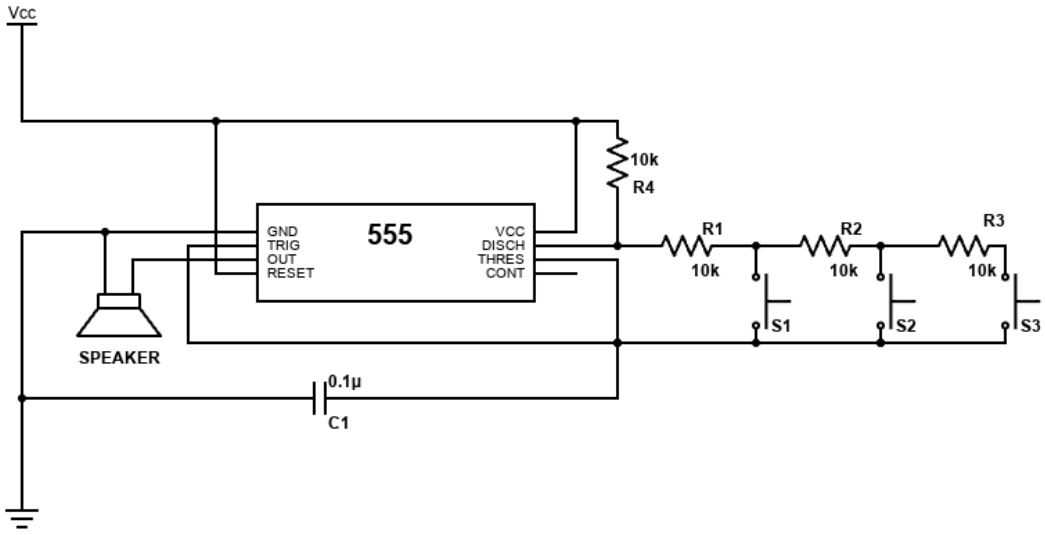


Figure 2: Example 555 Timer Schematic (Part 1)



Part 2:

Integrate an automated visual sequencer driven by the 555 timer's clock output into the 74HC595 8-bit shift register circuit. This stage scales down the operational frequencies to a human-visible pace using four LEDs and an interactive push-button input.

In Part 1, the 555 timer's output produced audible sound waves through a speaker; in Part 2, that exact same output now serves as the master clock signal (heartbeat) for the 74HC595 shift register, dictating the precise timing for when serial data is shifted and locked into the LEDs.

1. Connect the 555 timer to the 8-bit shift register's clock and logic controls.
2. Add a push button to register inputs to the shift register
3. Add your LEDs/resistors to the shift register
4. **MAKE SURE TO REPLACE THE 0.1 uF CAPACITOR WITH A 100 uF CAPACITOR** (Ensure that the anode is connected to the threshold pin and the cathode is connected to ground)

Adding this much larger capacitor will slow down each of the three frequencies enough to visually see data move across the 4 LEDs one-by-one. When testing, make sure to hold the input button for the 8-bit shift register and one of the three frequency buttons simultaneously until you hear one click (clock cycle). Once you feed one input into the first LED, continue holding the frequency button and release the input button for the 8-bit shift register. This will feed one data input into the sequence of 4 LEDs.

Reference the provided circuit schematics and pinout diagrams for the implementation of the 8-bit shift register.

Figure 3: 8 bit Shift Register Pinout

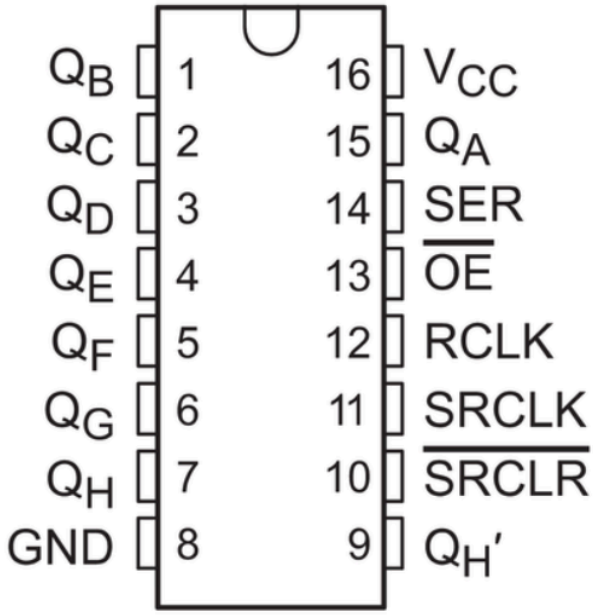
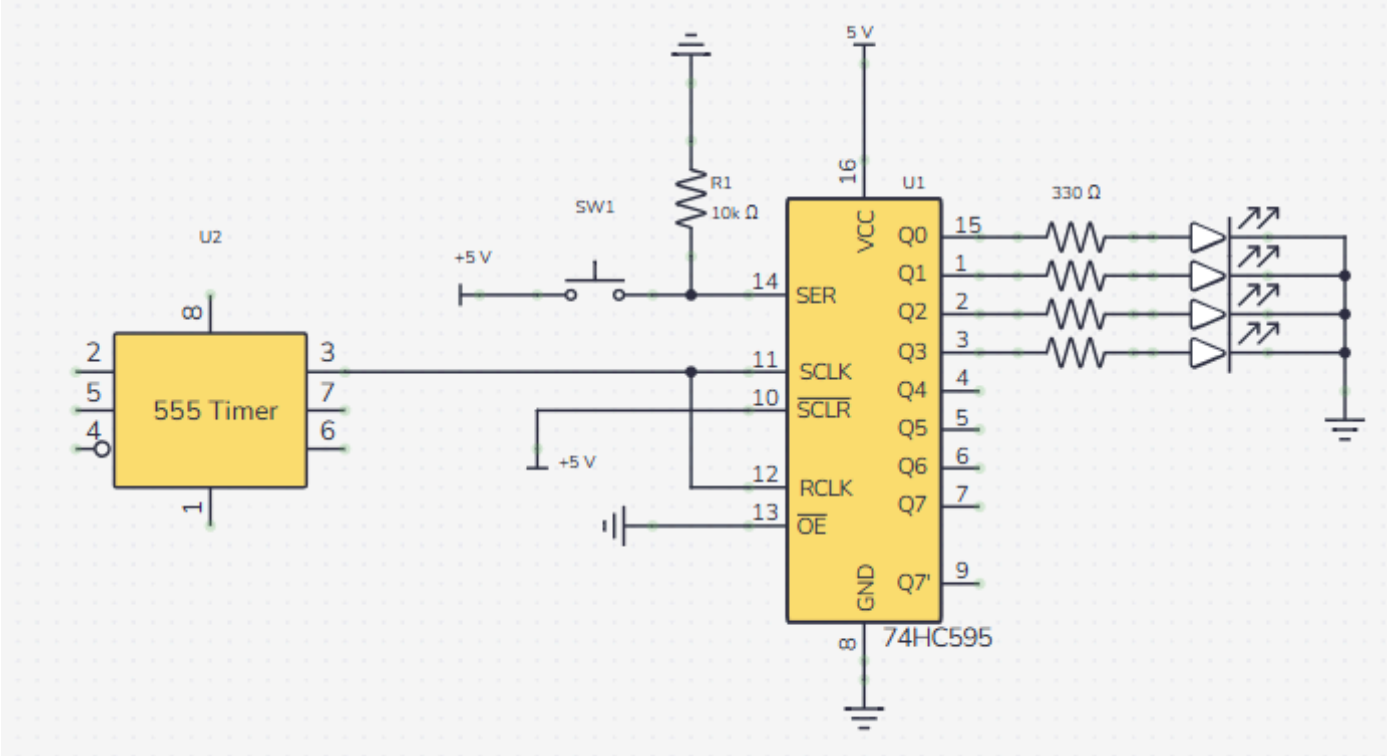


Figure 4: 8 bit Shift Register Schematic



What to Submit

Due Date: Tuesday, 11/10/26

Submit your completed **Part 1 and Part 2** circuit demonstration to **UF Check-In** as an unlisted YouTube video. Remember, if you ever have trouble with OPS, we are here to help! Feel free to reach out on the Slack discussion channel, contact OPS coordinators directly, or attend office hours.

Resources

[Lecture 3]

Hints:

Pay close attention to the position of resistors. Also note which wires need to be connected at junctions denoted by the red dots in the schematic provided.

Schematics:

Figure 5: Arduino Nano Pinout

